H

JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 1

CANDIDATE NAME

CLASS

BIOLOGY 8875/02

Paper 2

2 September 2013

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** the questions.

Section B
Answer **one** question.
Circle the question number of the question attempted.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
Section B	
5 / 6	
Total	

This document consists of **12** printed pages.

[Turn over

Section A

Answer **all** the questions in this section.

1 Fig. 1.1 shows a cell that has undergone semi-conservative replication and is in the process of undergoing telophase and cytokinesis.

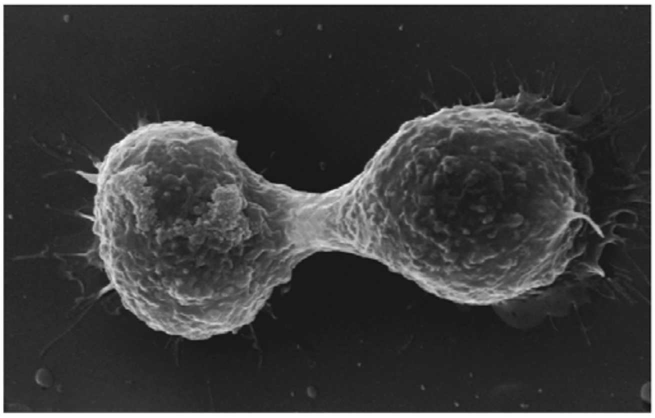


Fig. 1.1

(a) (i) State precisely when DNA replication occurs in the cell cycle. [1]

- **S phase of interphase;**

(ii) Outline the main features of semi-conservative DNA replication. [5]

1. **Unwinding of DNA double helix by helicase at origins of replication;**
2. **Breaking of hydrogen bonds between complementary base pairs, allowing separation of the two strands;**
3. **Attachment of single-strand DNA binding protein to parental DNA/ templates for stabilization;**
4. **Each parental strand serves as template for synthesis of new daughter strand;**
5. **Primase synthesizes RNA primer to initiate replication process;**
6. **DNA polymerase base pairs A=T, C≡G/free nucleotides with bases on template in specific, complementary manner;**
7. **Hydrogen bonds form between bases;**
8. **DNA polymerase catalyses the formation of phosphodiester bonds form between adjacent nucleotides of growing strand;**
9. **Synthesis of daughter strand occurs in 5' to 3' direction;**
10. **<u>Continuous</u> replication for daughter strand (synthesized in direction of unwinding of helix) / leading strand;**
11. **<u>Discontinuous</u> replication for daughter strand (synthesized in direction opposite of that of unwinding of helix) / Lagging strand/ small fragments called Okazaki fragments;**
12. **RNA primers removed and replaced by DNA nucleotides, by another DNA polymerase;**
13. **DNA ligase anneals sugar phosphate backbone/seals nicks between fragments via formation of phosphodiester bonds to form continuous strands;**

(b) Describe the events that occur during the stage shown in Fig. 1.1. [3]

1. **Daughter chromosomes reach opposite poles of the cell;**
2. **Chromosomes uncoil, (lengthen and become indistinct) to form chromatin;**
3. **Spindle fibres disintegrate;**
4. **Nuclear envelope reforms (around the chromatin in each daughter cell);**
5. **Nuclei formed;**
6. **Nucleoli reappear (in each daughter nucleus);**
7. **cleavage furrow begins to form / Cell starts to divide by constriction/furrowing/cleavage;**

(c) State how mitosis maintains genetic stability in an organism. [1]

1. **Semi-conservative DNA replication to form sister chromatids which are genetically identical to parent DNA;;**
   OR
2. **Separation of sister chromatids during anaphase leads to formation of genetically identical nuclei;;**
   OR
3. **Forms two nuclei/cell with the same number and type of chromosomes / same DNA as the parent;;**
   OR
4. **There is no crossing over;**
5. **there is no genetic recombination / there is no exchange of segments of non-sister chromatids of homologous chromosomes;**

   (R: Double DNA content, no genetic variation]

[Total: 10]

2 Fig. 2.1 is a diagram of an electron micrograph of part of a chloroplast showing the thylakoid membranes.

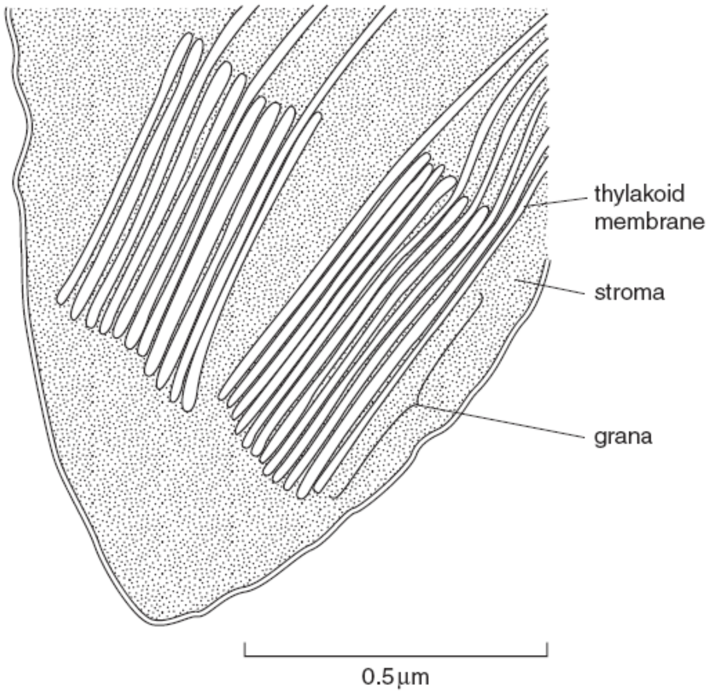


Fig. 2.1

(a) Describe the role of the thylakoid membrane in photosynthesis. [3]

1. **The thylakoid membrane is impermeable to protons;**
2. **and allow for the accumulation of H^+ in the thylakoid lumen to set up a proton gradient across the thylakoid membrane;**
3. **Provides a large surface area;**
4. **to hold the photosynthetic pigments, enzymes and electron carriers (any 2) needed for the light dependent reactions/cyclic and non-cyclic photophosphorylation;**
5. **Hold the photosynthetic pigments in a suitable position;**
6. **for absorbing the maximum amount of light energy;**
7. **Site of ATP synthesis;**
8. **(by ATP synthase) via chemiosmosis;**

(b) List three ways in which photophosphorylation differs from oxidative phosphorylation. [3]

Features	**No.**	**Photophosphorylation**	**No.**	**Oxidative phosphorylation**
Process	**1a**	In photosynthesis	**1b**	In aerobic respiration
Location	**2a**	Thylakoid membrane of chloroplast	**2b**	Inner membrane of mitochondrion
Involvement of light energy	**3a**	Light energy is required for photolysis / required to energise electrons in special chl a	**3b**	Light energy is not required
Source of energy for ATP synthesis	**4a**	Energy for synthesis of ATP comes from light	**4b**	Energy for synthesis of ATP comes from the oxidation of glucose
Electron donors	**5a**	Water is the electron donor in non-cyclic reaction	**5c**	NADH and $FADH_2$ are electron donors
	5b	Photosystem I is the electron donor in the cyclic reaction		
Electron acceptors	**6a**	$NADP^+$ is the final electron acceptor in the non-cyclic reaction	**6c**	Oxygen is the final electron acceptor
	6b	PSI is the electron acceptor in the cyclic reaction		
Establishing proton gradient for ATP synthesis	**7a**	Protons are pumped inwards	**7c**	Protons are pumped outwards
	7b	From stroma, across the thylakoid membrane, into the thylakoid space	**7d**	From matrix, across the inner mitochondria membrane, into the intermembrane space
H+ accumulation	**8a**	in thylakoid space	**8b**	In intermembrane space

Fig. 2.2 shows the results from two experiments carried out to investigate the effect of light intensity and carbon dioxide concentration on the rate of photosynthesis.

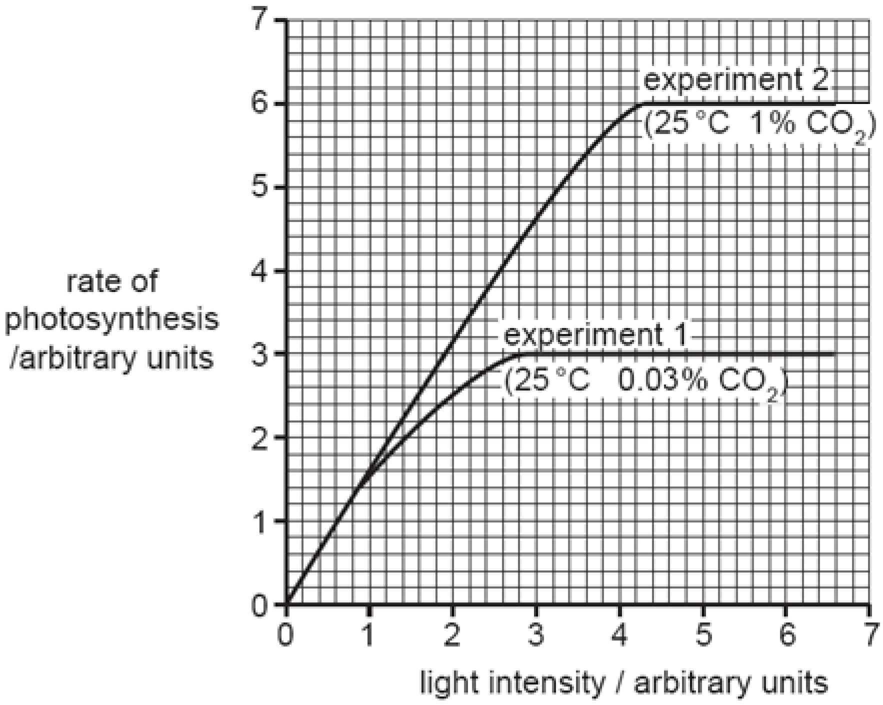


Fig. 2.2

(c) **(i)** Describe and explain the results shown in Fig. 2.2 for experiment 1. [3]

1. **As light intensity increases from 0 a.u. to 1 a.u., rate of photosynthesis increases (linearly/proportionally) from 0 a.u. to 1.6 a.u.;;**
2. **Light intensity is the (main) limiting factor;**
3. **As light intensity increases from 1 a.u. to 2.8 a.u., rate of photosynthesis increases (at a decreasing rate) from 1.6 a.u. to 3 a.u.;;**
4. **Light intensity is becoming less of a limiting factor, CO_2 concentration is starting to be limiting;**
5. **As light intensity increases from 2.8 a.u. to 6.6 a.u., rate of photosynthesis becomes constant/reaches a plateau at 3.0 a.u.;;**
6. **Light intensity is no longer a limiting factor and other factors e.g. CO_2 concentration is now the limiting factor;**

To further investigate the effects of carbon dioxide concentration on the Calvin cycle, the levels of ribulose bisphosphate (RuBP) and glycerate-3-phosphate (GP) were determined and the results shown in Fig. 2.3.

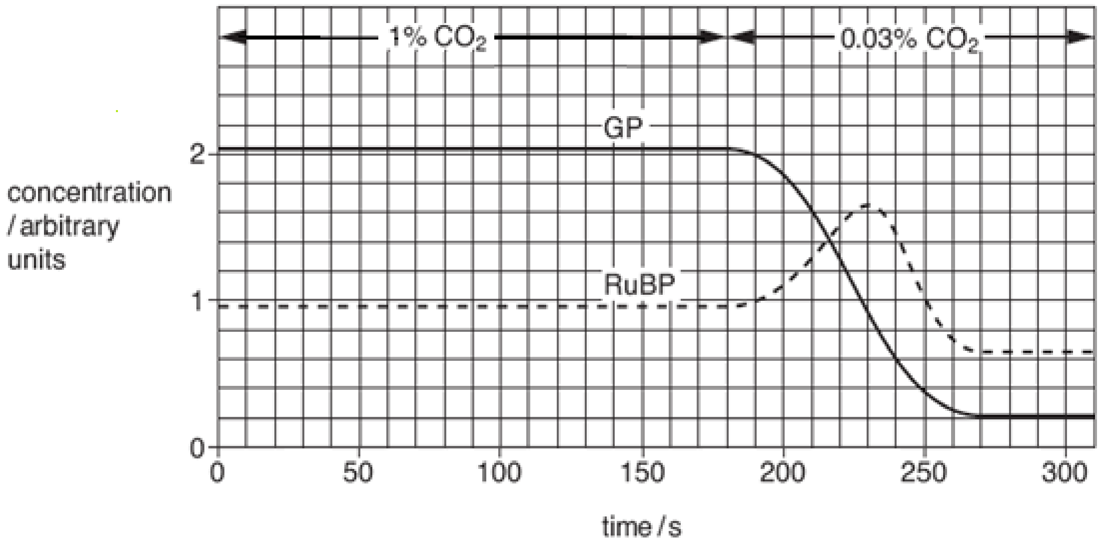


Fig. 2.3

(ii) Explain why the concentration of RuBP increases during the 50 seconds immediately after the carbon dioxide concentration is reduced to 0.03%. [2]

1. **RuBP is a carbon dioxide acceptor / fixes carbon dioxide/ combines with carbon dioxide to form GP (during carboxylation);**
2. **Reaction is catalysed by Rubisco;**
3. **GP converted to TP;**
4. **RuBP regenerated from TP;**
5. **Lower CO_2 concentration, less CO_2 available, less RuBP used;**
6. **RuBP builds up/ accumulates;**

[Total: 11]

3 The takahē, *Porphyrio hochstetteri*, is a flightless bird that is restricted to a small area of the South Island of New Zealand. It is one of only two remaining species of large, flightless, herbivorous birds from New Zealand. The other remaining species is found on the North Island of New Zealand. Their flighted ancestor came over from Australia millions of years ago.

The takahē was thought to be extinct, but a small population with low genetic variation was discovered in 1948 among the remote mountains of the South Island. The takahē is a grassland specialist and lives in alpine grassland dominated by tussock grasses as shown in Fig. 3.1.

Fig. 3.1

(a) Explain why it has been possible for flightless birds, such as the takahē, to evolve in New Zealand. [3]

1. **Flighted birds migrated to New Zealand;**
2. **(genetic) variation in the takahē's ability to fly due to (spontaneous) mutation;**
3. **Some birds were flightless while some were flighted;**
4. **Selection pressure – absence of mammals, abundant food / resources on ground / nesting sites available on the ground / few or no land predators;**
5. **As flight requires considerable energy;**
6. **Flightless birds with higher fitness are better adapted to the environment and will be at a selective advantage;**
7. **They will survive to maturity, mate / reproduce and pass down favourable alleles to their offspring;**
8. **Over time, change in allele frequency occurs, leading to evolution (and flightless birds predominates);**

(b) Suggest why low genetic variation in the population of takahē is undesirable. [1]

1. **Sudden changes in environment could result in widespread death of the species / species extinction / reduce chances of survival;;**

(c) To increase genetic variation in the population, conservationists selected some birds with more variation in their alleles and allowed them to interbreed. This is an example of artificial selection.

(i) Suggest one difference between artificial selection and natural selection. [1]

1. **Selection pressure by humans vs. environmental selection pressures;;**
2. **Genetic diversity lowered vs. genetic diversity remains high;;**
3. **Usually faster change in allele frequency vs. slower change in allele frequency;;**
4. **Selected feature not always to organism's advantage vs. selected feature always to organism's advantage;;**
5. **Inbreeding common vs. outbreeding common;;**

(ii) Describe two processes that led to increased genetic variation in the population. [2]

1. **Crossing over between non-sister chromatids of homologous chromosomes during prophase I of meiosis I;;**
2. **Independent assortment/segregation of homologous chromosomes during metaphase I/anaphase I;;**
3. **Independent assortment/segregation of chromatids during metaphase II/anaphase II;;**
4. **Random fusion of gametes during fertilisation;;**
5. **Gene mutation or chromosomal aberration;;**

Fig. 3.2 shows the relationship between some animals based on their anatomical homology by comparing the arrangement of the bones in their forelimbs which had a basic pentadactyl structure.

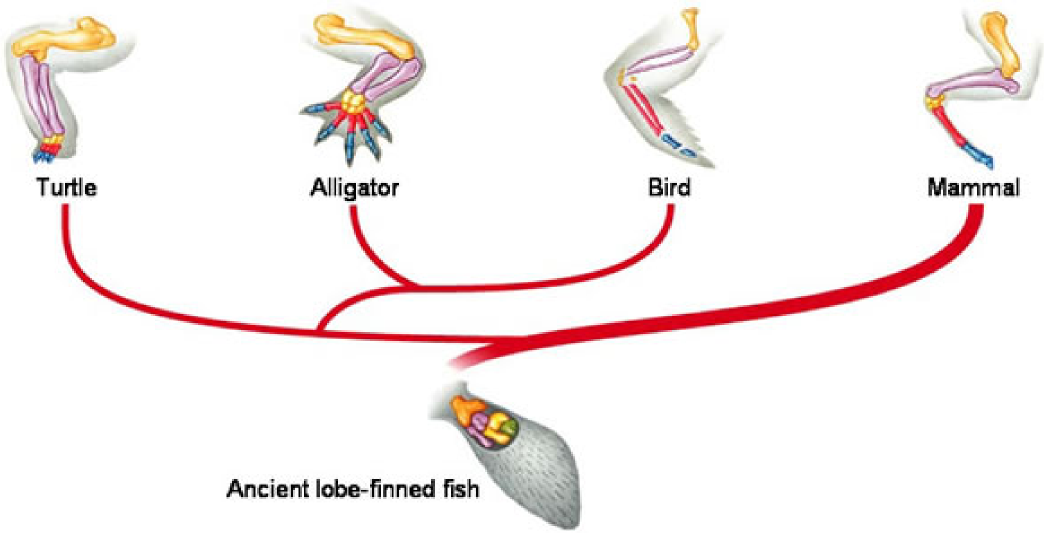


Fig. 3.2

(d) Explain how the anatomical homology in Fig. 3.2 provides evidence to support the theory of evolution. [3]

1. **Although the forelimbs had different functions;;**
2. **they basically have the same arrangement/similar structure;**
3. **which is a basic pentadactyl/ five digit forelimb structure;**
4. **Therefore they were likely to have originated/derived from a common ancestor;**
5. **Which is the ancient lobe-finned fish;**
6. **and had been modified over time/ descent with modification;;**

[Total: 10]

4 Insulin deficiency is a widespread problem in developed countries. To alleviate the symptoms, insulin injection is given to these patients. Insulin is manufactured on a large scale basis using genetic engineering techniques involving restriction enzymes and plasmids.

(a) Explain what is meant by a restriction enzyme. [2]

1. **Restriction enzyme recognises and binds to short sequences of bases of DNA (restriction sites);;**
2. **The restriction site is 4-6 base pairs in length;;**
3. **Catalyses the breaking of phosphodiester bonds between specific nucleotides / cut DNA at specific points;;**
4. **To produce restriction fragments with blunt/sticky ends;;**

Plasmids used in genetic engineering include pMOD2-Neo shown in Fig. 4.1. The pMOD2-Neo plasmid consists of 2871 bp and has an origin of replication (ori), a multiple cloning site (MCS) with restriction sites for 9 different restriction enzymes, a gene giving resistance to the antibiotic neomycin (neo) and a gene giving resistance to the antibiotic ampicillin (amp).

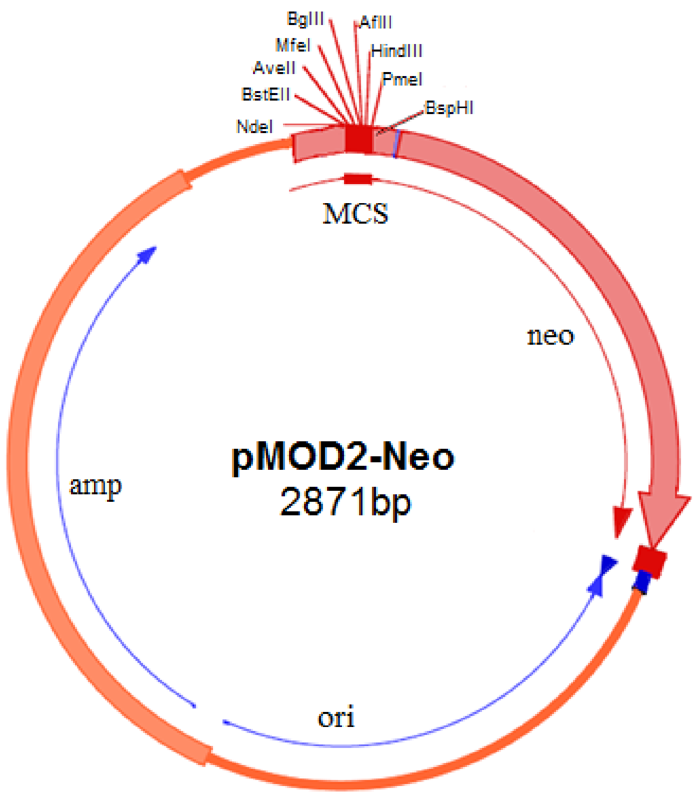


Fig. 4.1

(b) With reference to Fig. 4.1, describe how each region of the plasmid allows it to be used for cloning:

(i) Origin of replication [1]

1. Allows pMOD2-Neo/plasmid and foreign gene/GOI to replicate independently of bacterial genome within the host cell;;
2. To give multiple copies of the plasmid and gene within one bacterium;;

(ii) Ampicillin resistance gene [1]

1. (Transformed) bacteria containing pMOD2-Neo with the ampicillin resistance gene will be able to survive and grow on medium containing ampicillin;;
2. Allows for the selection of successfully transformed bacteria containing (both recombinant and non-recombinant) plasmid;;

R: Selection of transformed bacteria with recombinant plasmid, survival of bacteria not plasmids

(iii) Multiple cloning site [2]

1. Contains recognition sites for (9) different restriction enzymes. The choice of restriction enzymes allows pMOD2-Neo to be used to contain a wide range of foreign genes/allows a range of different genes to be cloned;;
2. Insertion of gene at the multiple cloning site leads to insertional inactivation of neomycin gene. This allows selection of bacteria containing recombinant plasmid;;

Scientists have developed another treatment for these patients, such as the use of stem cells which can be stimulated to develop into pancreatic cells that can synthesise and release insulin.

(c) State the normal functions of the stem cells used during the treatment. [2]

1. Can undergo differentiation giving rise to specialised cell types e.g. pancreatic cells;;
2. Capable of dividing indefinitely and producing copies of themselves;;

(d) Discuss the differences between adult stem cells and embryonic stem cells. [1]

1. Adult stem cells are multipotent and have the ability to differentiate into a limited range of cell types;;
2. whereas embryonic stem cells are pluripotent and have the ability to differentiate into almost any cell type to form whole organisms or type of cell in the body;;

[Total: 9]

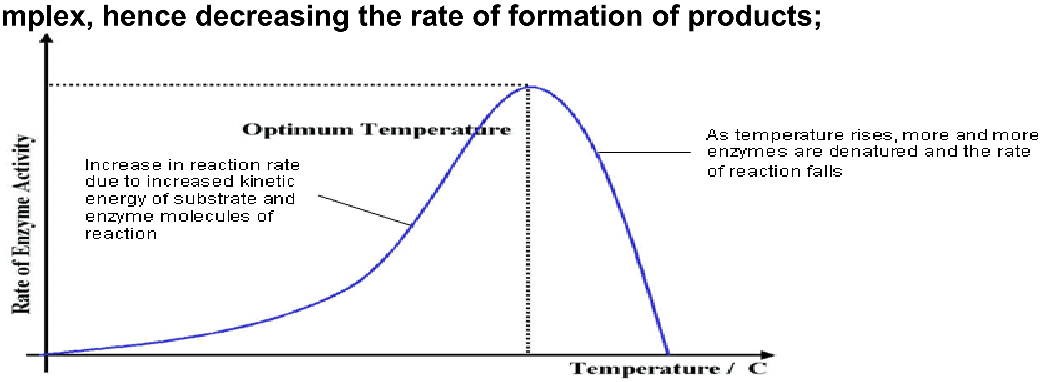
Section B

Answer EITHER 5 OR 6.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5 **(a)** Explain how temperature affects the rate of an enzyme-catalysed reaction. [8]

1. **At very low temperatures (near or below 0°C), enzyme activity is very low;**
2. **The enzyme is said to be inactivated;**
3. **due to possession of minimal kinetic energy/has low KE for collision with substrate(s)**
4. **As temperature increases to , before optimum temperature is reached, the rate of reaction increases;**
5. **This is because heating increases kinetic energy of the substrate(s) and enzyme;**
6. **hence increasing molecular motion of these molecules;**
7. **thus increasing the <u>frequency of successful collisions between the substrate(s) and enzyme</u>;**
8. **This increases the <u>rate of formation of E-S complexes</u>, thus increasing the rate of formation of products;**
9. **For most enzymes, Q_{10}= 2 i.e. the rate of an enzyme-controlled reaction is <u>doubled</u> for every rise of <u>10°C</u>.**
10. **At optimum temperature, maximum rate of reaction occurs;**
11. **At the optimum temperature, there is highest frequency of successful collisions between substrate(s) and enzyme/maximum rate of formation of E-S complexes;**
12. **Different enzymes function optimally at different temperatures;**
13. **As temperature rises beyond the optimum temperature, the rate of the reaction decreases;**
14. **despite substrate(s) and enzyme having increased kinetic energy and higher frequency of collisions;**
15. **This is because heat has broken the hydrogen bonds and hydrophobic interactions;**
16. **within the secondary and tertiary structures of the enzyme;**
17. **causing a loss of the (precise) 3D conformation of the enzyme's active site;**
18. **and a loss of the 3D conformation of the enzyme;**
19. **enzyme denaturation occurred;**
20. **Substrate(s) can no longer bind to the active site of the enzyme to form E-S complex, hence decreasing the rate of formation of products;**



(b) Outline the ways in which mutations can occur. [6]

M1. Factors that promotes rate of mutation – exposure to carcinogen, ultraviolet radiation, X-ray, AVP (any 2);
M2. Error occurring during replication/repair;

Gene mutation:
G1. Gene mutation is defined as the changes in the nucleotide base sequence of the genes.
G2. Substitution – replacement of one nucleotide base-pair with another base-pair in a gene;
G3. results in missense mutation/nonsence mutation/significant change in the encoded protein/silent mutation/no/little effect on the encoded protein.
G4. Insertion – addition of loss of one or more nucleotide base pairs in a gene;
G5. Deletion – loss of one or more nucleotide base pair in a gene;
G6. results in frameshift mutation, whereby codons are grouped differently;
G7. Inversion – a sequence of nucleotides becomes separated from the allele, it rejoins the original position but inverted;

Chromosomal mutation:
C1. Chromosomal mutation is defined as the changes in the nucleotide base sequence of the genes or the amount of chromosome.
C2. Structural alteration to chromosomes
C3. Translocation – a section of chromosome breaks off and become attached to another chromosome; (leading to a new combination of alleles)
C4. Duplication – (addition of genes) a section of a chromosome replicates (so that a set of gene loci is repeated) ;
C5. Inversion – a chromosome breaks at two locations and (the middle portion) flips through 180° before rejoining;
C6. Deletion – (loss of genes) a chromosome breaks at two points, the middle portion is displaced (with the two ends joining together); (A shorter chromosome containing fewer genes will form).

C7. Change in chromosome number
C8. e.g. Aneuploidy – addition or loss of one or more chromosomes (in the nucleus) or
C9. due to non-disjunction during meiosis/ fusion of a normal haploid gamete with a gamete carrying n-2, n-1, n+1 or n+2 chromosomes;
C10. Polyploidy - three or more times the haploid number of chromosomes;
C11. due to non-disjunction / the fusion of a diploid gamete with a normal haploid gamete giving a triploid/tetraploid nucleus

(c) Discuss the social implications of genetically modified organisms. [6]

<u>(a) Threat to human safety</u>

S1. **Ref. to transgenic food causing allergies with named examples (e.g. Bt toxin in Bt corn/genes in Golden rice coming from several organisms);;**

S2. **Ref. to <u>long term</u> unexpected/negative effect of transgenic food on human health with named examples;;**

S3. **Ref. to use of antibiotic resistance genes as selectable markers in vectors used for transforming plants and concerns in making bacterial species in human gut more resistant;;**

<u>(b) Threat to safety of the environment</u>

S4. **Ref. to cross-pollination/unintended transfer of transgenes from GM crops/crop-to-weed hybridization with named example (e.g. soy crop plants with herbicide-resistance);;**

S5. **Ref. to GM crops establishing as weeds/weeds becoming 'superweeds' and thus are invasive/reduces biodiversity/disrupts ecological balance;;**

S6. **Ref. to GM crops being toxic to non-target organisms with named example (e.g. Bt toxin affecting larvae of monarch butterflies);;**

S7. **Ref. to reduced numbers/extinction of non-target organism/disruption of food chain/disruption of ecological balance;;**

<u>(c) Issues pertaining to Access and Intellectual Property</u>

S8. **Ref. to why research companies seek patents (e.g. make profits/protect results of research);;**

S9. **Ref. to impact of patents (e.g. increase in price of seeds/ domination of world food production by few companies;;**

S10. **Ref. to biopiracy and how developed countries exploit the resources of developing countries;;**

<u>(d) Labelling is not mandatory in some countries</u>

S11. **Ref. to need for labelling GM food (e.g. religious/medical/dietary concerns);;**

S12. **Ref. to varying standards in labelling transgenic food with named examples (e.g. labelling threshold is 1% for Brazil and 5% for Japan/ labelling not mandatory in US);;**

S13. **Ref. to difficulties in maintaining standard of labelling due to pollen drift with named example (e.g. unauthorized corn in Mexico)**

<u>(e) Other social implications</u>

S14. **Ref. to advances being skewed to interests of developed countries/benefiting only rich countries;;**

S15. **Ref. to dominance of world food production by developed countries / increasing dependence of developing countries on industrialized nations;;**

S16. **Ref. to impact on international trade with named example (e.g. Europe being more hesitant in accepting GM products compared to US);;**

[½ mark for insufficient discussion of point]

[Total: 20]

6 **(a)** Describe the process of glycolysis and Krebs cycle. [10]

Glycolysis

1. **Glucose is phosphorylated by a molecule of ATP to form Glucose-6-phosphate;**
2. **Glucose-6-phosphate is isomerised to form Fructose-6-phosphate;**
3. **Fructose-6-phosphate is phosphorylated by a molecule of ATP to form Fructose-1,6-bisphosphate;**
4. **Fructose-1,6-bisphosphate is split to form 2 molecules of triose phosphate/ triose phosphate and dihydroxyacetone phosphate;**
5. **(Dihydroxyacetone phosphate is isomerised into the more stable triose phosphate);**
6. **For every TP, 2 molecules of Triose phosphate are phosphorylated by 2 molecules of inorganic phosphate;**
7. **and oxidised by 2 molecules of NAD^+ to form 2 molecules of glycerate bisphosphate;**
8. **2 molecules of NAD^+ are reduced to form two molecules of NADH;**
9. **2 molecules of glycerate bisphosphate are converted to form 2 molecules of Glycerate phosphate;**
10. **2 molecules of ATP are formed by substrate level phosphorylation (SLP);**
11. **2 molecules of glycerate phosphate are converted to form 2 molecules of pyruvate ;**
12. **4 molecules of ATP are formed by substrate level phosphorylation (SLP);**
13. **net gain of 2 ATP;**

(Award once for Pt 10 and Pt 12)

Krebs cycle

1. **Acetyl CoA (2C) and oxaloacetate (4C) are condensed to form citrate (6C);**
2. **Citrate (6C) undergoes oxidative decarboxylation to form α-ketoglutarate (5C);**
3. **A molecule of CO_2 is lost, and the remaining 5C compound is oxidised by NAD^+;**
4. **A molecule of NAD^+ is reduced to form NADH;**
5. **α-ketoglutarate (5C) undergoes oxidative decarboxylation to form succinate (4C);**
6. **A molecule of CO_2 is lost, and the remaining 4C compound is oxidised by NAD^+;**
7. **A molecule of NAD^+ is reduced to NADH;**
8. **A molecule of ATP is formed by substrate level phosphorylation;**
9. **Succinate (4C) undergoes oxidation to form fumarate (4C);**
10. **FAD is reduced by two molecules of hydrogen atoms to form $FADH_2$;**
11. **Fumarate (4C) is converted to form malate (4C);**
12. **Malate (4C) undergoes oxidation to regenerate oxaloacetate (4C);**
13. **A molecule of NAD^+ is reduced to NADH;**

(Award once for 1) Pt 3 and Pt 6; 2) Pt 4, Pt 7 and Pt 13)

(b) Describe how the eukaryotic gene can be incorporated into *E.coli*. [4]

1. **Desired gene is isolated (R: cut) using a restriction enzyme;**
2. **Cut the plasmid vector with the same restriction enzyme to get complementary sticky ends;**
3. **Mix the desired gene and plasmid vector in vitro;**
4. **allowing the sticky ends to reanneal in a specific, complementary manner/via complementary base pairing through the formation of hydrogen bonds between the bases;**
5. **DNA ligase is added to the mixture to seal the nicks between the fragments through the formation of phosphodiester bonds;**
6. **so that a recombinant plasmid is formed;**
7. **Add $CaCl_2$ to the mixture and subject the mixture to heat shock/electroporation;**
8. **so that transient pores formed on the bacteria cell membrane;**
9. **Allowing the recombinant plasmid to enter host *E.coli* cells via transformation;**

(c) Discuss the benefits of the Human Genome Project. . [6]

<u>Genetic Testing</u>

1. **The HGP has allowed for the discovery of genes/alleles associated with human diseases and allowed for better understanding of the genetic basis of diseases;;**
2. **This led to the development of genetic testing which allows for more accurate diagnosis of diseases, earlier detection of genetic dispositions to diseases (e.g. breast cancer, cystic fibrosis, Alzheimer's disease) and more accurate prediction of the onset of diseases;;**
3. **It also allows scientists to design drugs to target a specific gene/protein (i.e. rational drug design);;**

<u>Gene Therapy</u>

4. **The identification of genes associated with human genetic diseases and a better understanding of the genetic basis of disease may increase the potential to treat genetic and acquired diseases by gene therapy;;**
5. **Gene therapy allows defective genes to be either replaced by normal functional genes or supplemented with other genes to prevent a disease taking its course;;**

<u>Pharmacogenomics</u>

6. **The HGP allows scientists/doctors to know which genes/alleles affect an individual's response to a drug and how the genetic differences affect the way the drugs are metabolised in our bodies;;**
7. **This may eventually allow doctors to prescribe the right / effective drug in the right dosage for each individual;;**
8. **It is also theoretically possible to customise drugs/treatments to fit the patient's genome for greater efficacy as well as for avoiding dangerous side effects;;**
9. **Many customised drugs work by targeting specific proteins within our body. In customised drug design, knowledge of the gene sequence that codes for these specific proteins allows us to deduce 3D conformation of these proteins. This will allow us to design drugs that are complementary to the protein target;;**

<u>DNA forensics</u>

10. **The HGP allows for the discovery of genes/alleles/polymorphisms that can be used to:**
    (a) identify potential suspects whose DNA may match evidence left at crime scene, or clear innocent people of wrongdoings;;
    (b) identify mass casualties;;
    (c) establish paternity and other family relationships;;
    (d) identify protected/endangered species as an aid to wildlife officers as the evidence could be used to prosecute poachers;;

[1/2 mark for insufficient discussion of point]

[Total: 20]